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Evaluation of global leaf area index simulated by LM4+, the land component of KIOST-ESM2

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INTRODUCTION

Background: KIOST is developing KIOST-ESM2 for CMIP7 with GFDL's LM4+ (LM4 family) as its land component. LM4+ grows vegetation as size-structured cohorts (PPA), so leaf area, biomass and canopy structure evolve with climate.
Vegetation controls the land–atmosphere exchange of energy, water and carbon; a biased leaf area index (LAI) therefore feeds back on the coupled climate. Offline evaluations show that most land models keep leaves too long in autumn (Murray-Tortarolo et al., 2013); offline runs isolate the land model's own errors.

Key question: Does offline LM4+ reproduce the magnitude and the phenology of global LAI, and what causes the mismatch?

Objective 1: Benchmark 1982–2010 LAI against GIMMS LAI4g (pattern, magnitude, seasonal cycle).

Objective 2: Trace the autumn leaf-off delay to the phenology rule and test a day-length cue in pilot runs.

DATA & METHODS

Dataset	Setup / variables	Period / resolution
LM4+ offline (KIOST-ESM2)	PPA cohorts, 6 species, fixed species map, N cycle off, CO_2 fixed	1979–2023, C96 (≈1°) → 1°
WFDE5 forcing (Cucchi et al. 2020)	ERA5 bias-corrected with CRU/GPCC; T, q, wind, SW, LW, precip.	1979–2023, 0.5° hourly → C96
Initial land state	KIOST-ESM2 ≈400-yr coupled piControl + 1 cycle WFDE5 1981–2010	start 1979-01-01
GIMMS LAI4g (Cao et al. 2023)	AVHRR LAI, consolidated with MODIS	1982–2010, 15-day 1/12° → monthly 1°

Evaluation: 1982–2010 monthly climatology on a common 1° grid; MJJAS mean and bias; seasonal cycles in four IPCC AR6 regions. SOS/EOS = dates at which the interpolated monthly cycle crosses 50 % of its amplitude (White et al., 1997).
Pilot runs (year 2010 from the 1 Jan 2010 CTRL state): TC10 raises the leaf-off temperature threshold 5 → 10 °C; DL14+TC10 adds a day-length (< 14 h) leaf-off condition (code patch; larch and C3 grass).

RESULTS 1: GROWING-SEASON LAI

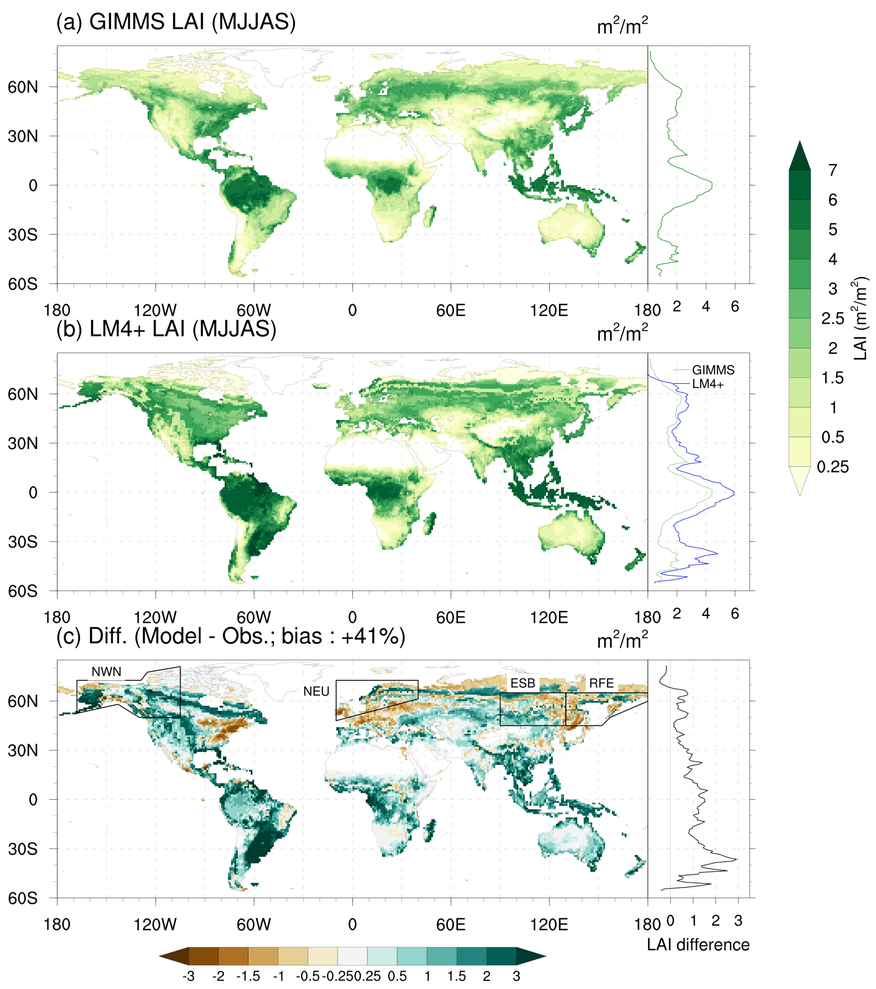


Figure 1. Growing-season (MJJAS) LAI, 1982–2010: (a) GIMMS LAI4g, (b) LM4+, (c) LM4+ − GIMMS. Side panels: zonal means. Polygons: the AR6 regions used in Figs. 2–3.

- The large-scale pattern is reproduced, but LAI is too high almost everywhere: global MJJAS mean 2.77 vs 1.97 (+41 %); annual +54 %.
- Largest excess over tropical and subtropical South America and Africa.
- **Boreal forest (50–70°N): the summer peak is only +15 % too high, yet the annual mean is +56 % too high — 87 % of the bias comes from the off-season (Sep–Apr). The extratropical excess is a timing problem.**

RESULTS 2: PHENOLOGY (OBS. AND MODEL)

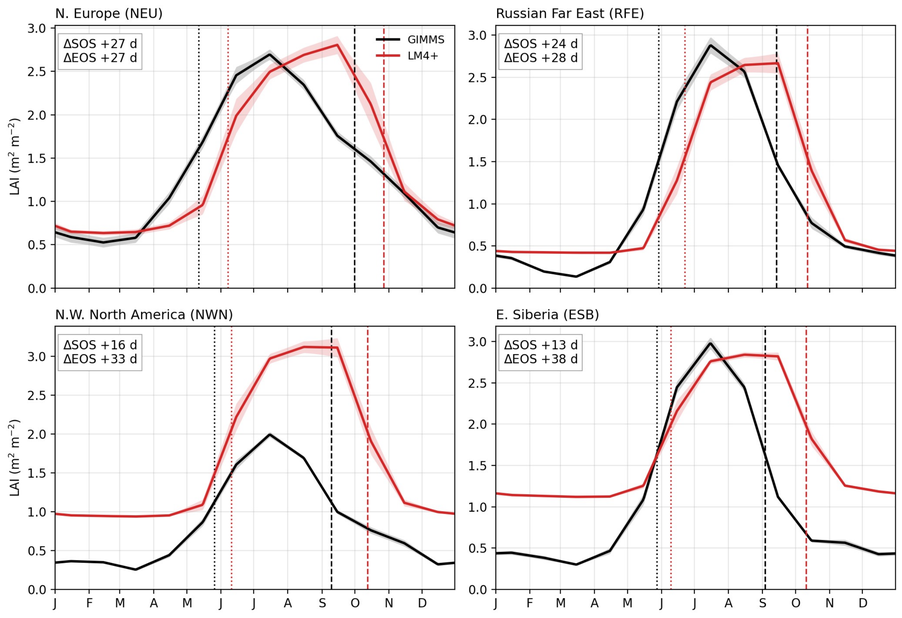


Figure 2. Climatological LAI seasonal cycle, 1982–2010, GIMMS vs LM4+ in four AR6 regions (shading: ±1 s.d. over 29 years); dotted / dashed lines: SOS / EOS at 50 % of the amplitude.

- **Leaf-off (EOS) is 27–38 days too late in every region (NEU +27, RFE +28, NWN +33, ESB +38 d) and in all 29 years; leaf-on is also late (+13–27 d).**
- Cause: leaves are dropped only when the 20-day smoothed canopy temperature falls below 5 °C (10 °C for broadleaf trees). Daily T < 5 °C is reached ≈20 d after the observed EOS and the smoothing adds ≈17 d: ESB observed 3 Sep → model 11 Oct.
- The observed EOS falls at a day length of 12.7–13.3 h in NWN, ESB and RFE: a day-length cue is missing in the model.

Phenology rule (Weng et al., 2015, Eq. A31)

$$GDD(t) = \sum_{i=1}^{t} \max[T_d(i) - 0\,^\circ\mathrm{C},\ 0] \qquad t = \text{days since the end of the last growing season}$$

$$T_{pheno}(t) = 0.95\,T_{pheno}(t-1) + 0.05\,T_d(t) \qquad \text{weighted mean daily temperature (memory} \approx 20\ \text{d)}$$

Leaf-on

$$GDD(t) > GDD_{crit} \quad \text{and} \quad T_{pheno}(t) > T_{crit}$$

Leaf-off

$$T_{pheno}(t) < T_{crit} \qquad \text{temperature is the only autumn cue}$$

$T_{crit} = 5\,^\circ$C (larch, C3 grass), $10\,^\circ$C (other deciduous) in the LM4+ configuration used here

RESULTS 3: MODIFIED PHENOLOGY RULES (PILOT)

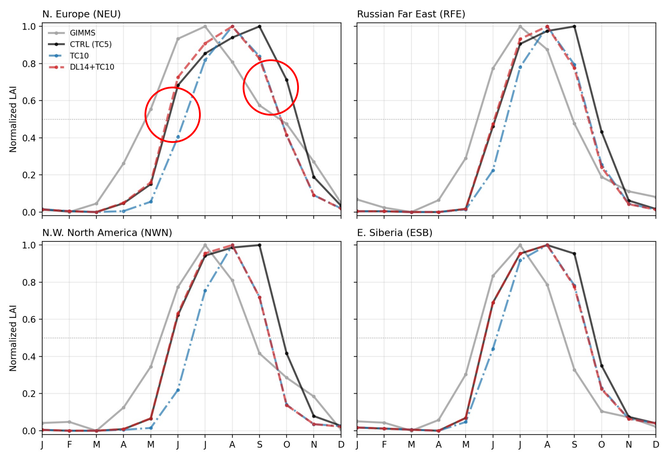


Figure 3. Normalized 2010 LAI cycle, (LAI − min)/(max − min), in the four regions with the largest gain: GIMMS, CTRL, TC10 and DL14+TC10.

TC10 (temperature only): autumn drop moves, but spring leaf-on is delayed.
DL14+TC10 removes the Sep–Oct excess by NEU 65 %, RFE 50 %, NWN 38 %, ESB 20 %, with spring unchanged.
50–70°N EOS bias: +33 → +16 d.
Sep–Oct fluxes change too (50–70°N, TC10): ET −4.5 %, GPP −11 %.

SUMMARY & CONCLUSIONS

- ✓ **Offline LM4+ reproduces the global LAI pattern but overestimates it (+41 % in MJJAS); in the extratropics the excess comes mainly from a 1–2 month delay of autumn leaf-off.**
- ✓ **The delay follows from a temperature-only phenology rule (5 °C threshold on a 20-day smoothed temperature); observations point to day length as the missing cue.**
- ✓ **A day-length leaf-off rule removes about half of the boreal EOS bias with no spring side effect; further threshold tests are under way.**
- ✓ Next: replant species from an observed map (larch is nearly absent in the current state), time-varying CO_2 and the C–N version, then coupled KIOST-ESM2 experiments.

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